

Software Requirements Specification (SRS)

Hostezy: Smart Hostel & Mess Management System

Date: 15th February 2026

Team Members

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1. Introduction

1.1. Purpose

This document describes the **software requirements** for the **Hostezy: Smart Hostel & Mess Management System**.

The purpose of this SRS is to provide a **clear and complete description** of the system's functional and non-functional requirements for developers, testers, faculty evaluators, and stakeholders.

1.2. Scope

The Smart Hostel & Mess Management System is a **web-based platform** designed to automate and manage hostel and mess operations efficiently.

- Student registration and authentication
- Hostel, room, and occupancy management
- Complaint and issue tracking
- Mess management with subscription-based access (weekly/monthly)
- Secure online payments
- Automated payment reminder before subscription expiry
- Admin and warden dashboards
- Notifications, reports, and analytics

1.3. Intended Audience

Stakeholders, developers, testers, project guide and evaluators, system administrators, and hostel operations personnel.

1.4. Product Features

- User authentication (Student, Admin, Warden)
- Student profile management
- Hostel, room, and allocation management
- Complaint registration and resolution

- Mess attendance and menu management
- Subscription-based mess access (weekly / monthly)
- Payment and fee management using Razorpay
- Automated payment reminders before subscription expiry
- Admin and warden dashboards
- Notification and communication system
- Reports and analytics

1.5. Non-Functional Requirements

- Performance: System responds within 2 seconds for standard operations
- Security & Privacy: JWT authentication, role-based access, HTTPS, secure payment gateway
- Maintainability: Modular and well-documented codebase
- Usability: Responsive UI for desktop and mobile
- Reliability: 99.9% system availability
- Scalability: Supports increasing number of users and hostels

1.6. Architecture Overview

- Frontend: React
- Backend: Node.js, Express.js
- Database: MongoDB
- Authentication: JWT
- Payment Gateway: Razorpay
- Hosting: Vercel

1.7. Future Enhancements

- Mobile application support
- AI-based mess demand prediction
- Biometric-based attendance
- Smart IoT-based room monitoring
- Integration with institute ERP

1.8. User Stories for Hostezy: Smart Hostel & Mess Management System

ID	User Role	Wants To	So That
US001	Student	register and login	access hostel services securely
US002	Student	view my room and hostel details	know my accommodation status
US003	Student	raise a complaint	report hostel-related issues
US004	Warden	view and resolve complaints	maintain hostel discipline
US005	Admin	manage hostels and rooms	control infrastructure
US006	Student	subscribe to mess weekly/monthly	access mess services
US007	Student	renew mess subscription online	continue mess access
US008	System	send payment reminder before expiry	avoid service interruption
US009	Student	pay fees online	complete transactions easily
US010	Student	view payment history	track my expenses
US011	Admin	manage mess plans	control pricing and duration
US012	Admin	generate reports	analyze hostel usage
US013	Warden	approve vacate requests	manage room availability
US014	Student	receive notifications	stay informed
US015	Admin	manage users and roles	maintain system control

1.9. Hostezy: Smart Hostel & Mess Management System - Project Plan

Overview

- Project Duration: Approximately 8–10 weeks
- Development Methodology: Agile (Scrum with weekly sprints)

Phase 1: Requirement Analysis & Planning (Week 1)

Deliverables:

- Finalized SRS
- Feature prioritization
- Technology stack decision
- Project roadmap

Tasks:

- Stakeholder discussion
- Finalization of user stories
- Project planning

Phase 2: Design (Week 2)

Deliverables:

- UI wireframes
- Database schema
- System architecture

Tasks:

- Dashboard design
- User flow design

Phase 3: Backend Development (Week 3–4)

Deliverables:

- REST APIs
- JWT authentication
- Database integration

Tasks:

- User, hostel, mess modules
- Payment integration
- Notification service

Phase 4: Frontend Development (Week 5–6)

Deliverables:

- Student portal
- Admin dashboard
- Warden dashboard

Tasks:

- API integration
- Responsive UI

Phase 5: Testing (Week 7)**Deliverables:**

- Test cases
- Bug fixes

Tasks:

- Functional testing
- Security testing

Phase 6: Deployment (Week 8)**Deliverables:**

- Cloud deployment
- Production configuration

Phase 7: Final Submission (Week 9)**Deliverables:**

- Final project demo
- Documentation

Tools & Technologies

Area	Tools / Stack
Frontend	React
Backend	Node.js, Express
Database	MongoDB
Payments	Razorpay
Auth	JWT
DevOps	GitHub
Design	Figma
Testing	Postman

Timeline Overview

Phase	Duration (Weeks)	Sprint(s)
Req. Analysis & Planning	1	Sprint 1
Design	1	Sprint 2
Backend Development	2	Sprint 3–4
Frontend Development	2	Sprint 5–6
Testing	1	Sprint 7
Deployment	1	Sprint 8
Final Submission	1	Sprint 9